

Week 1: The Nature of Strategic Interaction

Strategic Games, Dominant Strategies, and the Prisoner's Dilemma

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When your best decision depends on what someone else decides.

Return to the Pricing Decision

The Situation

Two competing firms choose prices simultaneously. Neither observes the other's current choice before deciding.

	Rival: Maintain	Rival: Cut
You: Maintain	(8, 8)	(2, 12)
You: Cut	(12, 2)	(4, 4)

Can we solve this without guessing what the rival will do?

Payoffs represent profits; your payoff is listed first.

Today's Learning Outcomes

By the end of class, you should be able to:

- 1 read and construct a two-player **payoff matrix**;
- 2 identify each player's **best responses**;
- 3 identify whether a player has a **dominant strategy**;
- 4 explain why individually rational choices may produce a poor collective outcome;
- 5 defend a business recommendation using explicit payoff assumptions.

Diagnose the Pricing Game

Players and Strategies

- **Players:** two competing firms
- **Strategies:** maintain price or cut price

Payoffs and Structure

- **Payoffs:** profits from each outcome
- **Structure:** one-shot and simultaneous

Simultaneous Does Not Mean the Same Clock Time

It means that neither firm observes the other's current choice before deciding.

How to Read a Payoff Matrix

	Maintain	Cut
Maintain	(8, 8)	(2, 12)
Cut	(12, 2)	(4, 4)

- The row player chooses a row.
- The column player chooses a column.
- The first number is the **row player's payoff**.
- The second number is the **column player's payoff**.
- Compare one player's payoffs at a time.

Payoffs may be actual profits or numbers representing how players rank the outcomes. Their interpretation must be stated.

Best Response: Hold the Rival's Choice Fixed

Definition

A player's **best response** is the action that gives that player the highest payoff for a particular action chosen by the rival.

If the rival maintains:

- Maintain gives you 8.
- Cut gives you 12.
- Your best response is **Cut**.

If the rival cuts:

- Maintain gives you 2.
- Cut gives you 4.
- Your best response is **Cut**.

Fix the rival's action. Then compare your own payoffs.

Dominant Strategy

Definition

A strategy is **strictly dominant** if it gives a higher payoff than every alternative, whatever the other player chooses.

- If the rival maintains, cutting gives $12 > 8$.
- If the rival cuts, cutting gives $4 > 2$.
- Therefore, **Cut** is your strictly dominant strategy.
- A player may have no dominant strategy in another game.

Best against every rival action = dominant strategy

Solving Both Firms' Decisions

	Rival: Maintain	Rival: Cut
You: Maintain	(8, 8)	(2, 12)
You: Cut	(12 , 2)	(4 , 4)

- Cutting is your best response whether the rival maintains or cuts.
- The rival faces the same comparisons, so cutting is also the rival's dominant strategy.
- The resulting outcome is mutual price cutting, with payoffs (4, 4).

Bold numbers mark best responses; the green box marks the resulting outcome.

The Prisoner's Dilemma

	Rival: Maintain	Rival: Cut
You: Maintain	(8, 8)	(2, 12)
You: Cut	(12, 2)	(4, 4)

- Each firm has an individual incentive to cut its price.
- Both firms therefore cut their prices, producing (4, 4).
- Yet both firms would earn more under mutual price maintenance: (8, 8).

The Strategic Dilemma

Individually sensible choices can produce an outcome that is worse for everyone.

Individual Incentives Versus Collective Performance

Strategic Stability

Each firm has an individual incentive to cut its price, regardless of the rival's action.

Collective Performance

Both firms would earn more if both maintained their prices.

Individually rational choices need not produce the best collective outcome.

Application 1: Food-Delivery Vouchers

Discussion Scenario

Two competing delivery platforms simultaneously decide whether to offer a costly weekend voucher. A voucher may attract users when the rival does not offer one, but matching vouchers reduce both platforms' margins.

Discuss the business logic:

- What can a unilateral voucher gain?
- What happens when both offer vouchers?
- What happens when neither does?

Build the model:

- Identify players and strategies.
- Rank the four outcomes.
- Construct a payoff matrix.
- Test for dominant strategies.

Does your matrix have a Prisoner's-Dilemma structure? Why?

Solving the Model Versus Checking the Model

Solve the Model

What follows if the stated strategies and payoffs are correct?

Identify best responses and determine whether a dominant strategy exists.

Check the Model

Are the strategies and payoff rankings credible in the actual business setting?

What evidence or changed assumption could reverse the conclusion?

A correct solution can still produce poor advice if the assumptions are wrong.

Application 2: Investment in AI Hardware

Discussion Scenario

Two competing firms simultaneously decide whether to make a large investment in AI hardware. Investing may improve future capability, but matching investments are costly and may leave their relative positions unchanged.

Discuss the incentives:

- What is gained by investing alone?
- What is lost by not matching a rival?
- How costly is mutual investment?

Construct and analyse:

- Define two strategies for each firm.
- Explain the four payoff rankings.
- Draw the payoff matrix.
- Identify any dominant strategy.

Comparison Question

Which assumption makes this game similar to, or different from, the voucher game?

Optional Extension: Airline Baggage Policy

Use If Time Allows, or Continue Next Class

Two airlines on an overlapping route simultaneously decide whether to charge separately for checked baggage. A fee generates ancillary revenue but may affect demand, brand position, and customer retention.

- 1 Define the players and their two strategies.
- 2 Explain how each airline ranks the four outcomes.
- 3 Construct a payoff matrix and test for dominant strategies.
- 4 Decide whether the game is a Prisoner's Dilemma.
- 5 Recommend an action and identify one assumption that could reverse it.

Checklist for the Two Application Matrices

Construct the Game

- Clear players and strategies
- Four outcomes explained before numbered
- Payoffs linked to business logic

Analyse the Game

- Compare one player's payoffs at a time
- Test every rival action for dominance
- State whether the PD label is justified

Final Check

Can another reader reconstruct the payoff ranking and identify the assumption driving your conclusion?

Application Debrief

- 1 Did the two applications produce the same strategic structure?
- 2 Which payoff assumption created or removed a dominant strategy?
- 3 When was the Prisoner's-Dilemma label justified?
- 4 What evidence would a manager need before acting?

Managerial Lesson

Solving a matrix is often easy. Constructing and defending the payoff ranking is the demanding part.

Week 1: Main Takeaways

- 1 A strategic game identifies the players, strategies, and payoffs from interacting choices.
- 2 A **best response** is the best action against a particular rival action.
- 3 A **dominant strategy** is best against every rival action.
- 4 A stable outcome may still be worse for everyone.
- 5 Business recommendations depend on defensible payoff assumptions.

Model, compare, predict, and then question the assumptions.

Exit Ticket

Answer in Two or Three Sentences

Give one business example of strategic interdependence. Identify the players and their decisions, and explain how one player's action changes the other's incentives.

Before Week 2

Complete the assigned pre-reading on sequential moves. Come ready to ask:

What changes when one player moves first and the other can observe that action?

Supplementary: Strict, Weak, and No Dominance

Case		Comparison of A with B	Conclusion
Strict dominance	dominance	A gives a higher payoff than B against every rival choice.	A is strictly dominant.
Weak dominance	dominance	A is never worse than B and is better against at least one rival choice.	A is weakly dominant.
No dominance		A is better against one rival choice, while B is better against another.	Neither dominates; further reasoning is needed.

Common Error

A strategy is not dominant merely because it produces the single highest payoff somewhere in the matrix.

Supplementary: Why Assumptions Matter

Possible Logic A

Fee revenue is large, customer switching is limited, and both airlines face similar incentives.

Charging a fee may become dominant and generate a Prisoner's-Dilemma-type outcome.

Possible Logic B

One airline has premium positioning, high-value customers, and a strong differentiation benefit from including baggage.

The airlines may have asymmetric payoffs and different best responses.

Same question, different defensible models, different recommendations.